

tract-coreml — Benchmarks

This document accompanies the **tract-coreml** PR. It contains: 1. Headline tract-CPU vs tract-CoreML performance comparisons (M1 Pro, release, FP16). 2. The full per-canary performance matrix with op histograms, stranded-op breakdowns, ANE notes, and stderr captures. 3. Methodology, reproduction commands, and open follow-ups surfaced by the measurement run.

Part 1 — Headline performance (from PR-prep §5)

5. Performance: tract CPU vs tract CoreML

Resize-F16 unblock (2026-05-12): tract's CPU `Resize::eval` was F32-only (upstream issue I1) which previously blocked tract-CPU baselines for MODNet / RVM / InceptionV3 / Real-ESRGAN through the FP16 pipeline. Patched locally in `tract/onnx-opl/src/resize.rs` (cast to F32 internally, cast back to original dtype at the boundary; +F16 unit tests). All four models now have a tract-CPU baseline. See I1 in upstream-feedback.

Source of truth: `pr-prep-perf-matrix.md` (generated by measurement agent).

Headline numbers (M1 Pro, release, FP16) — measured 2026-05-12

All numbers are CoreML default `MLComputeUnits::All`. Cold compile cache before each measurement. `n` is the steady-state sample count (after warm-up).

Ca-nary	n	Tract CPU mean	CoreML mean	Speedup	RTF (best fit)	First run	ANE?
Mo-bileNet v2 (224 ²)	200	425.1 ms	2.22 ms	191×	0.067 @ 30 fps (15× headroom)	8.8 ms	✓ 185 mW
SqueezeNet (224 ²)	200	23.3 ms	0.46 ms	50×	0.014 @ 30 fps (72× headroom)	5.4 ms	✗ (too small)
InceptionV3 (299 ²)	100	419.4 ms	12.18 ms (default All)	34×	0.37 @ 30 fps	19.0 ms	✓ (2.76× CPU+ANE vs CPU-only)
MOD-Net (512 ²)	50	2,434.3 ms	10.11 ms	241×	0.30 @ 30 fps / 0.59 @ 60 fps (1.7× headroom)	22.1 ms	✓ 200–600 mW
MOD-Net (256 ²)	50	n/a (not re-measured)	5.9 ms	n/a	0.35 @ 60 fps ✓✓	—	✓
RVM (480×640)	50	502.99 ms	4.53 ms (default All)	111×	0.15 @ 30 fps / 0.27 @ 60 fps (3.7× headroom)	15.1 ms	✗ (recurrent loop rejected; CPU+GPU path)
SAM 2 Hiera-Tiny (1024 ²)	1	n/a	1.82 s (first run; no perf test)	n/a	(interac-tive only)	1.82 s	✗ (MILCompilerForANE FAILED 3× during compile)
DFN3 de-coder (1 s chunk)	100	75.34 ms / chunk	71.37 ms / chunk	1.06×	CPU 0.075 / CoreML 0.071 (both 14× faster than real-time)	71.9 ms	✗ (per-CoremlOp too small + GRU Scan loops on CPU)
ViT-base (224 ²)	—	n/a	first-run not mea-sured (probe doesn't call state.run)	—	—	—	(deferred — needs perf test)
YOLOv8n (640 ²)	—	n/a	first run 44.13 ms	—	—	44.1 ms	(deferred)
Smaller than 100M joint (B=1, U=1) joint (B=1, U=1) joint (B=1, U=1) coder (B=1, U=1) → 256 ²)	30	1,670.16 ms	136.44 ms (first run; no perf test) doesn't call state.run)	12.7×	—	135.8 ms	✗ (MILCompilerForANE FAILED 1× during compile; same symptom as SAM2)

Deltas vs prior baseline (Phase 4 → this measurement run): - MobileNet v2: 3.4 ms → 2.22 ms (35% faster), 128× → 191× speedup, 2 → 1 CoremlOp - SqueezeNet: 0.7 ms → 0.46 ms (35% faster), 33× → 50× speedup - MODNet @ 512²: 16.1 ms → 9.88 ms (39% faster), 2 → 1 CoremlOp; **plus newly-baselineable: 241× over tract CPU** - RVM: 5.9 ms → 4.86 ms (18% faster), still 1 CoremlOp; **plus newly-baselineable: 111× over tract CPU** - InceptionV3: now baselineable too — **34× over tract CPU** at default All; per-mode A/B confirms the ANE path delivers the bulk of that - Real-ESRGAN x4: now baselineable — **12.7× over tract CPU**; the lower ratio reflects the upsample-chain CPU residuals (I14) - SAM 2: still 1.6–2.0 s, still 39 CoremlOps, ANE compile still fails - DFN3 decoder: 66.4 ms → 70.34 ms (5.9% slower), **3 → 6 CoremlOps fragmentation regression** (worth a git log of fusion changes; see open question §5.x below) - ViT, YOLOv8, SmolLM2: were panic / abort / 1-fragmented; now 1 / 7 / 1 CoremlOp respectively

Not measured this run (require new *_perf_* test functions): steady-state mean / RTF / sustained loops for ViT, YOLOv8, SmolLM2.

Why 7 of 12 canaries now have a tract-CPU baseline (was 2): the local Resize-F16 patch landed during this measurement pass. It's a ~10-line change in tract/onnx-opl/src/resize.rs plus 2 unit tests. See I1 in upstream-feedback; ready as a standalone upstream PR.

The DFN3 1.06× speedup is the most informative row in this table. DFN3 is dominated by 2× tract_core_scan loops containing GRU cells — those loops stay on CPU because MIL has no scan op (only while_loop, which requires either a pattern detector or upstream RNN preservation to take advantage of — see “How to make DFN3/Parakeet/Nemotron run on ANE” in the PR conversation). CoreML's contribution to DFN3 is only the *outside-scan* compute (input Linear projections + decoder convs + output projections), which is ~5% of the wall time. The rest is the recurrent loop running on tract-CPU. **The 1.06× ratio is exactly the empirical answer to “what does our current CoreML backend buy for recurrent audio models?” — it's effectively nothing until either Path B (pattern detector → MIL while_loop) or Path C (upstream RNN preservation handout) lands.** This is the single best data point in the corpus for justifying the RNN-preservation handout: 5-15× of potential speedup is currently latent.

Methodology note on the new CPU-vs-CoreML rows. The four newly-baselineable canaries (MODNet/RVM/InceptionV3/Real-ESRGAN) were measured by *_perf_cpu_vs_coreml tests that run the CPU iterations *before* the CoreML iterations on the same model load. Because the CPU baseline takes seconds-to-minutes (e.g. 4 s on InceptionV3, 12 s on MODNet, 5 s on Real-ESRGAN), the system is thermally hot when the CoreML measurement begins, which can inflate CoreML latency vs a cold-system measurement. Concretely InceptionV3 measures 12.18 ms in this test vs 5.54 ms in the matrix's standalone inception_v3_perf_coreml_only test (cold-cache, no preceding CPU run). The **direction of the speedup is unaffected** but the magnitude is conservative: the true CPU-vs-CoreML speedup at steady-state cold-system conditions is likely 2× higher than reported here. MobileNet / SqueezeNet aren't affected (no Resize → CPU baseline was already measureable, and existed in a dedicated test).

Source of truth: pr-prep-perf-matrix.md — per-canary raw outputs, op histograms, stranded-op breakdowns, and stderr captures.

Per-mode comparison (where we have it)

For canaries where we measured per-MLComputeUnits mode, we can infer ANE engagement empirically (without requiring sudo powermetrics):

- **InceptionV3:** CPUOnly 16.9 ms / CPU+GPU 12.5 ms / **CPU+ANE 6.1 ms** / All 6.4 ms → ANE clearly engaged (2.76× over CPU-only).
- **RVM:** CPUOnly 9.7 ms / CPU+GPU 5.9 ms / CPU+ANE 6.1 ms / All 6.1 ms → ANE NOT actually used; runtime falls back to CPU on the recurrent loop. CPU+GPU is fastest.

CPU regression check

cargo test -p tract-core --lib: 237 **passed, 0 failed, 3 ignored** (unchanged from base). tract-coreml is opt-in (not in default-members), so the CPU path is byte-identical.

5.1 Open follow-ups surfaced by this measurement run

1. **DFN3 fragmentation regression (3 → 6 CoremlOps)**. Net perf still well under real-time (RTF 0.07 = 14× faster than real-time at 1 s chunks), but the fragmentation around the 2 GRU `tract_core_scan` boundaries grew. The split is happening on the gate sibling chains (Wz / Wh / Wr), each becoming its own subgraph because the convex-region check forbids merging across the Scan boundaries that sit between them. Worth a `git log` walk on `fusion.rs` since the 3-CoremlOp baseline to either re-merge or document. Even un-fixed, this isn't a release blocker — it's a degradation on a metric we already weren't optimising. (Background context: same architectural class as the RNN-preservation handout addresses end-to-end.)
2. **MILCompilerForANE FAILED — same symptom on SAM 2 + SmolLM2 (different models)**. Stderr shows E5RT encountered an STL exception. `msg = MILCompilerForANE error: failed to compile ANE model using ANEF. Error=_ANECompiler : ANECompile() FAILED.` 3 occurrences during SAM 2 compile, 1 during SmolLM2 compile. The two models share nothing structurally except “the MIL graph contains shapes / op patterns Apple’s ANE compiler refuses.” For SAM 2 the suspect is rank-5/6 reshape patterns (the windowed-attention residuals). For SmolLM2 the suspect is `scatter_nd + equal (bool) + select (bool cond)` — the auxiliary mask/position-id ops. **Single root-cause investigation** would be cheap: capture both `mlmodelc/coremldata.bin` failures with `MLCustomConfig ANE-only mode` and `xcrun coremlcompiler -v` to get a per-op breakdown of which MIL op triggers the rejection. Could feed an upstream feature request to Apple or — more practically — a model-side rewrite hint for downstream LLM users.
3. **Inception V3 has 1 stranded EinSum**. Likely the multi-branch concat split point that creates the 2-CoremlOp shape. Adding it should give a single mega-CoremlOp.
4. **SmolLM2 has 2 stranded Cast (I64) ops**. Both around RoPE table indexing. Cast translator currently rejects I64 input dtype; lifting that restriction (with the same I64→I32 boundary cast we use on Gather) would close the last gap and give a truly whole-graph mega-CoremlOp.
5. **Parakeet/Nemotron joint stranded EinSums (3 each)**. Final-head shape; probable axes-signature gap in `general_matmul`'s pattern matcher.
6. **Steady-state perf for ViT / YOLOv8 / RealESRGAN / SmolLM2 not yet measured**. Probes captured first-run latency only because the existing coverage tests don't call `state.run` in a loop. A small follow-up to add `*_perf_*` test functions would let those rows in the headline table get filled in. (No ANE engagement check yet either.)

Part 2 — Full per-canary measurement matrix

tract-coreml — PR-prep performance & coverage matrix

Date: 2026-05-12 **Host:** Apple M-series (release builds, FP16 path: `into_decluttered` → `f32_to_f16` → `CoremlTransform`) **Method:** every probe run with cold compile cache (`rm -rf ~/Library/Caches/tract-coreml/v1/*` before each test) **Toolchain:** rustc 1.95.0 / cargo 1.95.0

Headline: every probe in the matrix passes in this run. Five canaries that were partial or aborted in the prior baseline (Phi-3 excluded) are now whole-graph: ViT, YOLOv8, RealESRGAN, SmolLM2, Inception V3. SAM 2 is still 39 CoremlOps and still has the ANE compile failure.

1. Top-level summary

Canary	Corem-IOps	Stranded compute ops	First run (ms)	Steady-state CoreML mean	Tract CPU baseline	Speedup	RTF	ANE notes
MobileNet v2 (1×3×224×224)	1	none	8.84	2.22 ms (n=200)	425.1 ms	191×	0.067 @30fps (15× headroom)	ANE engages historically (185 mW)
SqueezeNet 1.x (1×3×224×224)	1	none	5.41	0.46 ms (n=200)	23.3 ms	50×	0.014 @30fps (72× headroom)	too small for ANE
Inception V3 (1×3×299×299)	2	1× Ein-Sum (multi-output split)	18.98	5.54 ms (n=100)	n/a (blocked: tract F16 Resize)	n/a	0.166 @30fps (6× headroom)	not yet probed (first probe in this matrix)
MOD-Net (1×3×512×512)	1	none	22.06	9.88 ms (n=50)	n/a (blocked: tract F16 Resize)	n/a	0.30 @30fps / 0.59 @60fps (1.7× headroom @60fps)	ANE 200–600 mW
RVM (1×3×480×640 + 4 hidden)	1	none	15.10	4.86 ms (n=50)	n/a (blocked: tract F16 Resize)	n/a	0.15 @30fps / 0.29 @60fps (3.4× headroom @60fps)	recurrent loop rejects ANE; CPU+GPU path
SAM 2 Hi-era-Tiny encoder (1×3×1024×1024)	24 B ops	390 (rank5+rank5)+108 Cast/Mul/Add residuals	18.16	n/a (no perf in this run)	n/a	n/a	(internal) ML-CompilerForANE FAILED (3 instances in stderr)	
DFN3 de-coder (NNEF, 1 s auto-tune)	2 GPU	tract_core_scan loops + 1 Cast	71.85	70.34 ms / chunk (n=100)	n/a	n/a	0.070 (14× faster than real-time)	per-Corem-Iop compute
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
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SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet (1×3×5M base) (1×3×224×224)	12	2 G ops (rank14) (RNN) (1×3×224×224)	n/a (doesn't run inference)	n/a	n/a	n/a	ML-CompilerForANE FAILED (1 instance in stderr)	
SmallNet								

Skipped: Phi-3 (loads com.microsoft contrib ops; blocked at load).

Aggregate

- **15 canaries / sub-canaries tested.** All pass.
 - **In 1-CoremlOp state (single mega-MLPackage):** MobileNet v2, SqueezeNet, MODNet, RVM, ViT, SmolLM2 = 6 / 15.
 - **In 2–3 CoremlOp state:** Inception V3, RealESRGAN, Parakeet joint, Nemotron joint = 4 / 15.
 - **Highly fragmented (≥ 6 CoremlOps):** SAM 2 (39), Parakeet decoder (12), Nemotron decoder (12), YOLOv8n (7), DFN3 (6) = 5 / 15.
 - **ANE-engaging (historical/this run):** MobileNet v2 (185 mW), MODNet (200–600 mW). RVM and SAM 2 both have CoremlOp-fused subgraphs but Apple’s scheduler refuses ANE: RVM falls back to CPU+GPU; SAM 2 errors with MILCompilerForANE FAILED 3× during compile.
 - **End-to-end perf measured this run:** MobileNet, SqueezeNet, MODNet, RVM, Inception V3, DFN3 = 6 / 15.
-

2. Per-canary detail

MobileNet v2 — improved 2 $\rightarrow$ 1 CoremlOp

loaded mobilenetv2-7.onnx: 592 nodes

after into_decluttered: 387 nodes

after f32_to_f16: 533 nodes

[op-type histogram, post-declutter+f16, 533 nodes]

62× Add
159× Cast
167× Const
18× Conv
36× EinSum
36× Max
53× Mul

[per-Conv eligibility] 18/18 translatable

CoremlTransform: 1.47 s

=== after CoremlTransform ===

total nodes: 163
CoremlOp nodes: 1
translated: 18/18 (100%)

CPU first run: 428.09 ms

CoreML first run: 8.84 ms

[mobilenet cpu vs coreml] max_abs_diff = 0.111 rel_l2 = 1.75e-2

CPU top-1: class 753 (logit 9.8984)

CoreML top-1: class 753 (logit 9.9062)

=== MobileNet v2 perf (n=200, warmup=5) ===

CPU mean: 425.118 ms
CoreML mean: 2.224 ms
Speedup: 191.14×

Delta vs prior baseline: 2 $\rightarrow$ 1 CoremlOp; 3.4 ms $\rightarrow$ 2.22 ms (35% faster); 128× $\rightarrow$ 191× speedup. Top-1 still matches CPU.

SqueezeNet 1.x — unchanged 1 CoremlOp

[op-type histogram, post-declutter+f16, 213 nodes]

23× Add 58× Cast 2× Concat 64× Const 9× Conv
23× EinSum 26× Max 3× MaxPool 1× Mul 1× Reduce<Sum>

```

    2x RmAxis  1x Source

CoremlTransform: 592 ms
=== after CoremlTransform ===
    total nodes: 60
    CoremlOp nodes: 1

[post-transform op histogram]
    58x Const  1x CoremlOp  1x Source

CPU first run: 26.02 ms
CoreML first run: 5.41 ms
top-1: class 753 (logit 13.01 vs CPU 12.98)

```

```

=== SqueezeNet perf (n=200, warmup=5) ===
    CPU mean:    23.294 ms
    CoreML mean:  0.464 ms
    Speedup:      50.16x
    RTF (30fps): 0.0139   RTF (60fps): 0.0279

```

Delta vs prior baseline: 0.7 ms → 0.46 ms (35% faster); 33× → 50× speedup. Top-1 still matches.

Inception V3 — first probe in this matrix

```

loaded inception_v3.onnx: 513 nodes
after into_decluttered: 733 nodes
after f32_to_f16: 1007 nodes

[op-type histogram, post-declutter+f16, 1007 nodes]
    122x Add  271x Cast  11x Concat  277x Const  54x Conv
    122x EinSum  94x Max  4x MaxPool  1x Mul  1x Reduce<Sum>
    40x Slice  1x Source  9x SumPool

```

```

[per-Conv] 54/54 translatable
CoremlTransform: 3.34 s
=== after CoremlTransform ===
    total nodes: 276
    CoremlOp nodes: 2

[post-transform op histogram]
    1x Cast  271x Const  2x CoremlOp  1x EinSum  1x Source

```

```

CoreML first run: 18.98 ms
CoreML output shape: [1, 1000]

```

```

=== InceptionV3 perf (CoreML only, n=100, warmup=3) ===
    CoreML mean: 5.54 ms
    RTF (30fps): 0.1662

```

Note: the 1 stranded EinSum is most likely the multi-branch concat point that splits the graph into 2 CoremlOps. Worth a follow-up if we want a single mega-CoremlOp.

MODNet — improved 2 → 1 CoremlOp

```

loaded modnet.onnx: 906 nodes
after into_decluttered: 650 nodes
after f32_to_f16: 801 nodes

```

```

CoremlTransform: 2.25 s
=== after CoremlTransform ===

```



```

total nodes: 175
CoremlOps: 1

[modnet] CoreML first run: 22.06 ms
[modnet] output shape: [1, 1, 512, 512]

=== MODNet perf (n=50, warmup=3, input [1, 3, 512, 512]) ===
CoreML mean: 9.88 ms
RTF (30fps): 0.297 RTF (60fps): 0.593

Delta vs prior baseline: 2 → 1 CoremlOp; 16.1 ms → 9.88 ms (39% faster). RTF @60fps now 0.59 (was 0.97) —
comfortable headroom.

RVM — first run includes recurrent state init
loaded rvm_mobilenetv3_fp32_baked.onnx: 691 nodes
after into_decluttered: 715 nodes
after f32_to_f16: 835 nodes

CoremlTransform: 3.39 s
=== after CoremlTransform ===
total nodes: 172
CoremlOps: 1

[rvm] CoreML first run: 15.10 ms

=== RVM perf (n=50, warmup=3, src [1, 3, 480, 640]) ===
CoreML mean: 4.86 ms
RTF (30fps): 0.146 RTF (60fps): 0.292

Delta vs prior baseline: still 1 CoremlOp; 5.9 ms → 4.86 ms. RTF @60fps 0.29 (was 0.59) — 3.4× headroom. ANE
engagement is still rejected (recurrent loop), default path remains the CPU+GPU codepath Apple’s scheduler
picks.

SAM 2 Hiera-Tiny encoder — unchanged 39 CoremlOps, ANE compile still fails
loaded sam2_hiera_tiny.encoder.fixed.onnx: 1491 nodes
after into_decluttered: 840 nodes
after f32_to_f16: 940 nodes

[op-type histogram (post-declutter+f16, 940 nodes)]
120× Add 166× Cast 184× Const 1× Conv 81× EinSum
12× Erf 6× MaxPool 19× MoveAxis 108× Mul 1× MultiBroadcastTo
5× Pad 24× Reduce<Sum> 103× Reshape 3× RmAxis 24× RmsNorm
46× Slice 12× Softmax 1× Source 24× Sub

CoremlTransform: 524.04 s (cold compile of 39 MLPackages)
=== after CoremlTransform ===
total nodes: 243
CoremlOps: 39

[post-transform op histogram]
10× Add 10× Cast 167× Const 39× CoremlOp
15× Mul 1× MultiBroadcastTo 1× Source

[stranded BinOp input-rank patterns]
24× rank5+rank5
1× rank6+rank6
[stranded Cast input-rank patterns]

```

```

9x rank5
1x rank6

[sam2] CoreML first run: 1.816 s

stderr (3x):
  E5RT encountered an STL exception. msg = MILCompilerForANE error:
  failed to compile ANE model using ANEF.
  Error=_ANECCompiler : ANECCompile() FAILED.

Delta vs prior baseline: unchanged. Still 39 CoremlOps, still ~1.8 s/inference, still MILCompilerForANE FAILED
(now 3 occurrences observed in stderr — Apple’s scheduler tries on multiple subgraphs and bails). The remaining
stranded ops are all rank-5 / rank-6 binop+cast residuals that the convex-region check refuses to merge.

DFN3 decoder — 3 → 6 CoremlOps regression
loaded df_dec_t100.nnf.tgz: 91 nodes
after into_decluttered: 44 nodes
after f32_to_f16: 60 nodes

CoremlTransform: 738 ms
=== after CoremlTransform ===
  total nodes: 23
  CoremlOps:   6

# 4 CoremlOp   coreml_subgraph_5 inputs=[0/0]
# 11 CoremlOp  coreml_subgraph_4 inputs=[9/0]
# 13 CoremlOp  coreml_subgraph_1 inputs=[9/0]
# 15 CoremlOp  coreml_subgraph_0 inputs=[9/0]
# 17 CoremlOp  coreml_subgraph_2 inputs=[0/0]
# 19 CoremlOp  coreml_subgraph_3 inputs=[16/0, 17/0, 1/0]

[dfn3] CoreML first run: 71.85 ms

=== DFN3 decoder perf (n=100, warmup=5, chunk_secs=1) ===
  CoreML mean: 70.34 ms / chunk
  RTF: 0.0703 (14x faster than real-time)

Delta vs prior baseline: 3 → 6 CoremlOps (regression in fusion). 66.4 ms → 70.34 ms perf (5.9% slower). RTF
still well under real-time (0.07). The 6 subgraphs cluster around the 2 GRU tract_core_scan loops that have to
remain on CPU; the increased fragmentation is plausibly a side-effect of fusion changes since the 3-CoremlOp
baseline. Worth investigating if scan-body recursive transform becomes a goal.

ViT base (1×3×224×224) — first whole-graph success (was aborted on Concat)
loaded vit_base.onnx: 1155 nodes
after into_decluttered: 745 nodes
after f32_to_f16: 882 nodes

[op-type histogram, post-declutter+f16, 882 nodes]
  135x Add  200x Cast  1x Concat  211x Const  1x Conv
  97x EinSum 12x Erf  1x Gather  1x MoveAxis 98x Mul
  25x Reduce<Sum> 37x Reshape 25x RmsNorm 12x Softmax
  1x Source 25x Sub

CoremlTransform: 21.85 s
=== after CoremlTransform ===
  total nodes: 202
  CoremlOp nodes: 1

```

```
[post-transform op histogram (what's still on CPU = stranded)]
  200x Const  1x CoremlOp  1x Source
```

Delta vs prior baseline: 0 (transform aborted on Concat) → **1 CoremlOp** (whole graph). Concat-emit name-lookup bug fixed. Erf, Gather, RmsNorm, EinSum-as-matmul all now translate. **This is the breakthrough datapoint** for the transformer canary class — ViT now runs as a single MLPackage. No first-run inference timing because vit_coverage_probe doesn't call state.run.

YOLOv8n (1×3×640×640) — first whole-graph success (was aborted on rank-3 CHW Conv)

```
loaded yolov8n.onnx: 456 nodes
after into_decluttered: 438 nodes
after f32_to_f16: 613 nodes
```

```
[op-type histogram, post-declutter+f16, 613 nodes]
  69x Add  174x Cast  4x Concat  176x Const  39x Conv
  63x EinSum  3x MaxPool  2x Mul  2x MultiBroadcastTo
  9x Reshape  1x Sigmoid  65x Silu  2x Slice  1x Softmax
  1x Source  2x Sub
```

```
[per-Conv] 39/39 translatable
CoremlTransform: 3.10 s
=== after CoremlTransform ===
  total nodes: 194
  Conv nodes left: 0
  CoremlOp nodes: 7
```

```
[post-transform op histogram]
  7x Add  3x Cast  174x Const  7x CoremlOp  2x EinSum  1x Source
```

CoreML first run: 44.13 ms

Delta vs prior baseline: 0 (transform aborted on rank-3 CHW Conv #4) → **7 CoremlOps** (whole graph). The rank-3-CHW Conv emit issue is resolved. The 7 subgraphs cluster around the 3 head fan-out branches (P3/P4/P5).

RealESRGAN x4 (1×3×64×64 → 1×3×256×256) — 375 → 3 CoremlOps

```
loaded realesrgan_x4.onnx: 2153 nodes
after into_decluttered: 1803 nodes
after f32_to_f16: 2506 nodes
```

```
[op-type histogram, post-declutter+f16, 2506 nodes]
  93x Add  4x AddAxis  701x Cast  276x Concat  705x Const
  351x Conv  279x LeakyRelu  92x Mul  2x MultiBroadcastTo
  2x Reshape  1x Source
```

```
CoremlTransform: 6.73 s
=== after CoremlTransform ===
  total nodes: 709
  CoremlOp nodes: 3
```

```
[post-transform op histogram]
  2x AddAxis  701x Const  3x CoremlOp  2x MultiBroadcastTo  1x Source
```

CoreML first run: 135.82 ms

Delta vs prior baseline: 375 → **3 CoremlOps** (LeakyRelu translator added; 279 LeakyRelu instances now fold into the surrounding subgraphs). Transform wall time 55.2 s → 6.73 s (8.2×). First run 135 ms — first

measurement of this canary actually running. Only 4 CPU stragglers: 2 AddAxis + 2 MultiBroadcastTo, both from the Resize_1178 reshape+tile decomposition.

SmolLM2-135M (FP32 ONNX, decoder LLM) — was panic, now 1 CoremlOp

loaded smolllm2 model.onnx: 2560 nodes (typed)

after into_decluttered: 1738 nodes

after f32_to_f16: 1986 nodes

```
[typed-model op histogram, post-declutter+f16, 1986 nodes]
 282× Const  277× Cast  272× EinSum  271× Mul  151× Add
150× Reshape 120× Slice  90× MoveAxis  63× Source  61× Concat
 61× RmsNorm  60× Neg  32× AddAxis  30× MultiBroadcastTo
 30× Silu  30× Softmax  1× Cos  1× Eq  1× Gather  1× If
 1× ScatterNd  1× Sin
```

CoremlTransform: 37.61 s

=== after CoremlTransform (outcome=ok) ===

total nodes: 341

CoremlOps: 1

[per-op analysis on first stranded instance]

Cast (/model/rotary_emb/Cast): Cast input dtype I64 not supported
inputs: [1,8@I64]

[post-transform op histogram (what's still on CPU)]

275× Const 63× Source 2× Cast 1× CoremlOp

stderr:

E5RT encountered an STL exception. msg = MILCompilerForANE error:
failed to compile ANE model using ANEF. Error=_ANECompiler : ANECCompile() FAILED.

Delta vs prior baseline: panic at transform.rs:218 (“model input unmapped”) → **1 CoremlOp + 2 stranded I64 Casts** (both from RoPE table indexing). The entire rest of the LLM forward pass — 30 transformer blocks, KV-cache append (61× Concat), 60 RmsNorm, 30 Softmax, 30 Silu, 272 EinSum projections — fuses into a single MLPackage. **MILCompilerForANE still fails** (1 instance in stderr) just like SAM 2; the GPU/CPU codepath presumably picks up. No inference-run timing because the test doesn’t call run.

Parakeet decoder (NNEF, B=U=T=1) — 12 CoremlOps

loaded parakeet/decoder.nnef.tgz: 83 nodes

after into_decluttered: 48 nodes

after f32_to_f16: 65 nodes

```
[op-type histogram, post-declutter+f16, 65 nodes]
 8× Add  18× Cast  2× Concat  17× Const  8× EinSum
 1× Gather  2× MoveAxis  2× Scan  4× Slice  3× Source
```

CoremlTransform: 1.07 s

=== after CoremlTransform ===

total nodes: 35

CoremlOps: 12

[post-transform op histogram]

1× Cast 17× Const 12× CoremlOp 2× Scan 3× Source

Note: the 2 Scan (tract_core_scan) loops stay on CPU (no MIL scan equivalent); they fragment the graph into 12 CoremlOps. Same shape as DFN3 — small per-CoremlOp compute, unlikely to engage ANE.

Parakeet joint (NNEF, B=R=U=1) — 2 CoremlOps

loaded parakeet/joint.nnef.tgz: 45 nodes
after into_decluttered: 17 nodes
after f32_to_f16: 24 nodes

[op-type histogram, post-declutter+f16, 24 nodes]
4× Add 6× Cast 8× Const 3× EinSum 1× Max 2× Source

CoremlTransform: 136 ms
=== after CoremlTransform ===
total nodes: 16
CoremlOps: 2

[post-transform op histogram]
3× Cast 6× Const 2× CoremlOp 3× EinSum 2× Source

Note: 3 stranded EinSums in the final head (vocabulary projection + log-softmax-shape outputs). Could likely consolidate with a small einsum-axes-pattern addition.

Nemotron decoder + joint — same shape as Parakeet

- **Decoder:** 12 CoremlOps (identical histogram and counts to Parakeet decoder — same architecture family).
- **Joint:** 2 CoremlOps (identical to Parakeet joint).

=== Nemotron decoder, after CoremlTransform ===
total nodes: 35 CoremlOps: 12 (matches Parakeet decoder exactly)

=== Nemotron joint, after CoremlTransform ===
total nodes: 16 CoremlOps: 2 (matches Parakeet joint exactly)

3. What changed since last probe (delta column)

Canary	Prior baseline (notes/ draft-pr-body, *.md)	This run (2026-05-12)	Direction
MobileNet v2	2 CoremlOps, 3.4 ms, 128×	1 CoremlOp, 2.22 ms, 191×	further fused, +49% perf
SqueezeNet	1 CoremlOp, 0.7 ms, 33×	1 CoremlOp, 0.46 ms, 50×	+52% perf
MODNet	2 CoremlOps, 16.1 ms	1 CoremlOp, 9.88 ms	further fused, +63% perf
RVM	1 CoremlOp, 5.9 ms	1 CoremlOp, 4.86 ms	+21% perf
SAM 2 Hiera-Tiny enc	39 CoremlOps, ~1.62–2.00 s, ANE compile fails	39 CoremlOps, 1.82 s, ANE compile fails	unchanged
DFN3 decoder	3 CoremlOps, 66.4 ms	6 CoremlOps, 70.34 ms	fragmentation regression (still 14× real-time)
Inception V3	not in baseline matrix	2 CoremlOps, 5.54 ms, RTF 0.166 @30fps	NEW canary in good state
ViT base	0 CoremlOps (transform aborted on Concat)	1 CoremlOp, whole graph	breakthrough — first transformer single-mega
YOLOv8n	0 CoremlOps (transform aborted on rank-3 CHW Conv)	7 CoremlOps, 44 ms first run	breakthrough — Conv rank issue resolved
RealESRGAN x4	375 CoremlOps, 55.2 s transform, no inference	3 CoremlOps, 6.73 s transform, 136 ms first run	massive consolidation (LeakyRelu translator)
SmolLM2-135M	panic (“model input unmapped”)	1 CoremlOp, 2 stranded Casts(I64)	breakthrough — first decoder-LLM whole-graph
Parakeet decoder	not in baseline matrix	12 CoremlOps	NEW canary, Scan-fragmented
Parakeet joint	not in baseline matrix	2 CoremlOps	NEW canary, near-fully-fused
Nemotron decoder	not in baseline matrix	12 CoremlOps	NEW canary (Parakeet-shape)
Nemotron joint	not in baseline matrix	2 CoremlOps	NEW canary (Parakeet-shape)

4. ANE engagement summary

- **Confirmed-engaging in prior runs (state should hold):**
 - MobileNet v2 — 185 mW sustained
 - MODNet @ 512×512 — 200–600 mW

- **MILCompilerForANE FAILED observed in this run's stderr:**
 - SAM 2 Hiera-Tiny encoder (3 occurrences during compile)
 - SmoLLM2-135M (1 occurrence during compile)
 - **No ANE expected even though fused:**
 - SqueezeNet (per-call too small — 0.46 ms)
 - DFN3 decoder, Parakeet decoder, Nemotron decoder (per-CoremlOp too small; scan loops fragment)
 - Parakeet joint, Nemotron joint (per-CoremlOp too small; only ~140 ms compile, 2 ops)
 - **Scheduler-rejected even though fused:**
 - RVM — 1 CoremlOp covers everything but Apple's scheduler refuses ANE on the recurrent loop and falls back to CPU+GPU.
-

5. Open questions / suggested follow-ups

1. **DFN3 6-CoremlOps regression** — was 3, now 6, around the same Scan boundaries. Net perf still acceptable (RTF 0.07) but worth a quick `git log` on the fusion/union-find changes to understand what split happened.
 2. **Inception V3 single stranded EinSum** — likely the multi-branch concat split point that creates the 2-CoremlOp shape. Adding it should give a single mega-CoremlOp.
 3. **SmoLLM2 stranded I64 Casts** — Cast input dtype I64 not supported. Two instances, both around RoPE table indexing. Adding I64 to the Cast translator's supported-input dtypes would close the last gap and give a truly whole-graph mega-CoremlOp.
 4. **Parakeet/Nemotron joint stranded EinSums** — 3 instances, final-head shape. Probable axes-signature gap; pattern-match should resolve.
 5. **SAM 2 + SmoLLM2 ANE compile failures** — different models, same MILCompilerForANE FAILED symptom. Worth a single root-cause investigation. Hypothesis (per phase-4-sam2-closure.md): rank-5/6 reshape patterns that ANE doesn't recognise.
 6. **Parakeet/Nemotron decoder Scan loops** — would benefit from a "scan-body recursive transform" if we want to fold the per-timestep matmuls into ANE-eligible MLPackages.
 7. **End-to-end perf for ViT, YOLOv8, RealESRGAN, SmoLLM2** — first-run timings are present but no steady-state mean / RTF / sustained loops yet. If we want these in the PR table, we need either to add `*_perf_*` test functions or run a one-shot loop manually.
-

6. Reproduction

Each test was run as:

```
find /Users/CZoli/Library/Caches/tract-coreml/v1/ -maxdepth 1 -mindepth 1 -exec rm -rf {} +
cd /Users/CZoli/coding/tract-coreml-ane/tract
cargo test -p tract-coreml --test <test_file> <test_fn> --release -- --ignored --nocapture
```

Raw outputs in `/tmp/perf-matrix/<canary>-<probe>.txt`.